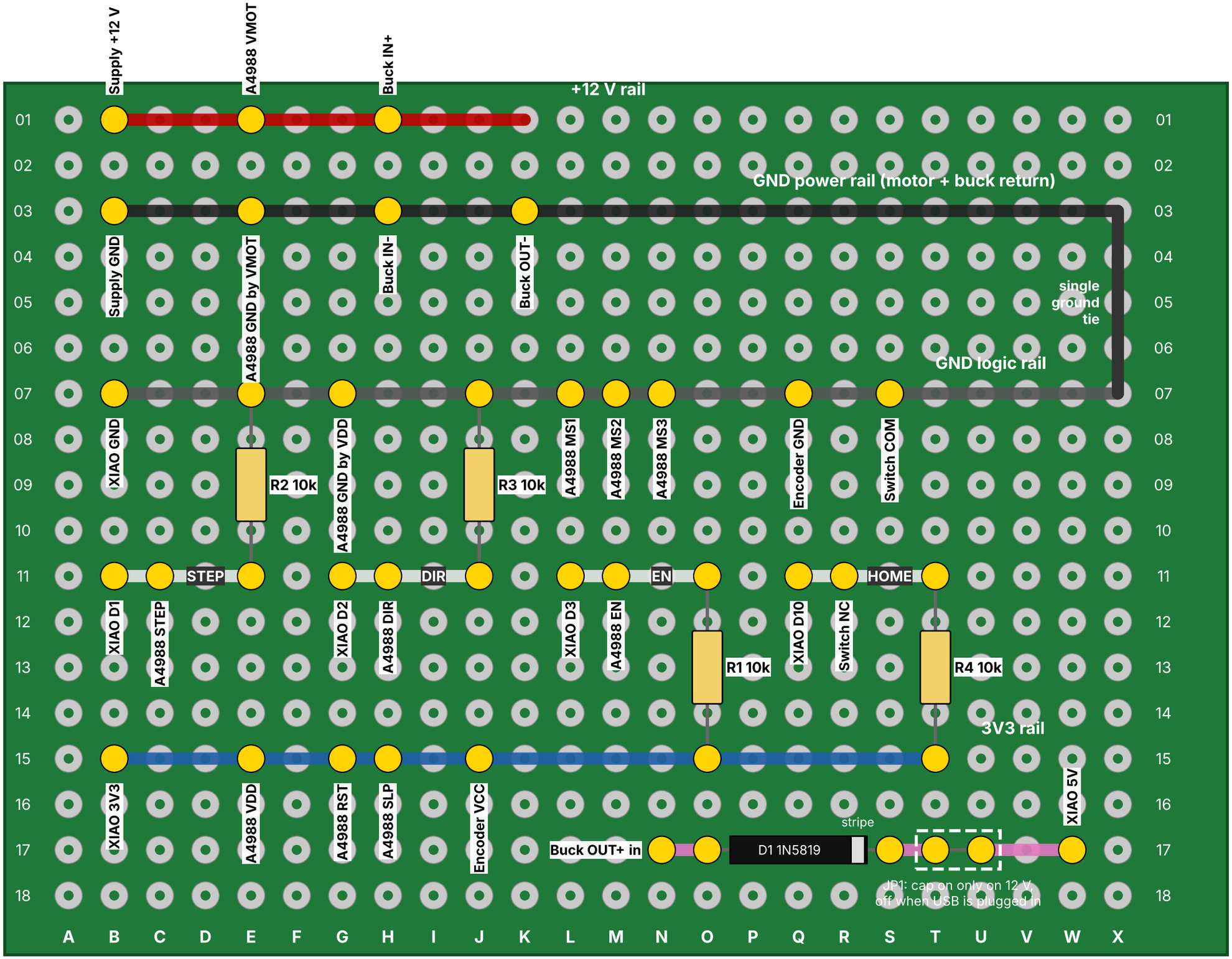


COMPONENT SIDE VIEW of a 5 × 7 cm perfboard (24 × 18 holes). Thick lines = bare wire / solder run on the bottom (rails and islands). Yellow = a hole you solder: a module wire (white label) or a part lead.

NOT on this board: XIAO D4 → encoder SDA, XIAO D5 → encoder SCL, A4988 1A/1B/2A/2B → motor, and C1 100 uF soldered straight across the A4988 VMOT and GND pins (+ to VMOT).